

COMPUTER VISION-BASED SMART TRAFFIC MANAGEMENT WITH A SMART RETRACTABLE SPEED BUMP AND A MOVABLE LANE DIVIDER

One camera measures the road in km/h — and the signals, the lane divider and a retractable speed bump all act on that one measurement, while an ambulance can flatten the bump over its own radio link

School of Computing · Asia Pacific University of Technology & Innovation · Group Design Project · Final Submission

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27.1
fps end to end
YOLO11-m @ 1280 px FP16

±5%
speed accuracy
calibrated vs. ground truth

0
night false alarms
down from 29% of all traffic

45° / 90°
bump raised / flat
calibrated against the plate

831/833
automated checks pass
7 suites, reproducible

21,526
vehicles logged
across 69 recorded sessions

Complete System Block Diagram

— one video feed becomes one measurement, and every actuator on the prototype is driven from it

five stages, four boards, two radios

1- VIDEO SOURCE

2- MEASUREMENT (CV node, laptop GPU)

3- DECISION

4- RELAY

5- ACTUATORS (physical prototype)

LOCAL INPUTS · INDEPENDENT EMERGENCY PATH

— the bump reads its own sensors and its own radio, so it still flattens for an ambulance with the laptop switched off

EMERGENCY VEHICLE
ambulance · police · fire

ARDUINO NANO
transmitter

SX1278 TX
433 MHz

command 'E'

SX1278 RX
433 MHz

HC-SR04
vehicle within 20 cm

PUSH BUTTON
congestion mode

ESP32 HUB
USB · COM6 · 7-field line

STREAMLIT DASHBOARD
HTTP 8502 · 9 pages · Excel

SIGNAL UNO
two signal heads · status LCD

LANE UNO
two divider servos · LED strip · I²C LCD

BUMP UNO
retractable bump servo · 45° / 90°

three local inputs

Background & Problem Statement

why this exists

Urban junctions still run on **fixed-time plans**. A timer holds a green over an empty approach and a red over a queue, because the junction cannot see the road it governs. Where sensing does exist it is usually an **inductive loop** cut into the carriageway — costly to install, blind to what kind of vehicle passed, and unable to report a speed.

Traffic calming is blunter still: a **fixed bump slows everything that crosses it**, including the ambulance behind, and lane allocation fixed in paint leaves a road full one way and empty the other right through the peak.

- Nothing on the road measures the road.** Signal timing, lane allocation and calming are all set open-loop, from a plan rather than from the traffic present.
- Calming is indiscriminate, and the parts do not talk:** a permanent bump penalises the one class of traffic that must not be delayed, and the signal, the divider and the bump are separate installations sharing no measurement.

Problem statement.

Urban traffic control is driven by plans instead of measurements: signals run on timers, lanes are fixed in paint, and speed bumps penalise emergency vehicles as heavily as speeding ones — because nothing on the road measures the traffic and shares it with the devices that ask on it.

Aim & Measurable Objectives

what counts as done

Aim.

To design, build and validate a camera-driven smart traffic prototype that measures real traffic in engineering units and uses that one measurement to drive adaptive signals, a movable lane divider and a retractable speed bump with emergency-vehicle priority.

- O1 - Measure speed.** Report per-vehicle speed in km/h from a single camera to within ±5% of ground truth once calibrated.
- O2 - Quantify congestion.** Produce one index on 0-1, updated **every second**, that falls monotonically as mean speed falls, over **≥10,000** samples.
- O3 - Adapt the signal.** Drive real signal hardware from that index with a **committed 5 s yellow** and a **≥3 s minimum dwell**, and no phase flicker when congestion flips every 0.5 s.
- O4 - Recognise emergency vehicles.** Grant priority only to a vehicle that is both liveried and flashing, with **zero false alarms** on night footage.
- O5 - Retract the bump.** Raise it for a vehicle detected within **20 cm** and flatten it on a **valid LoRa command**, under a **fixed emergency > congestion > automatic** priority.
- O6 - Reallocate lanes.** Move the divider left / right / normal on command and show system state on an LED strip and an LCD, verified over **7 commands**.
- O7 - Prove it.** Log every second to CSV and XLSX, and validate the software with **more than 800 automated checks**.

Justification for the Project

why it is worth building

- The problem is lethal and expensive.** The WHO puts road-traffic deaths at **1.19 million a year**, and congestion costs cities working hours and fuel daily.
- Minutes decide outcomes.** Survival after out-of-hospital cardiac arrest falls by roughly **7-10% per minute** before treatment: a bump in front of an ambulance is not a small inconvenience.
- Existing sensing is too expensive to spread.** A lamp means excavation, a lane closure and a fixed installation per approach. This prototype costs **RM 485** in parts and reuses cameras many junctions already have.
- Smart calming exists, but not at this price.** Commercial road plates prove the principle hydraulically, nothing in that class is emergency-aware at a scale a council could trial on one street — and this build still produced a reusable archive of **21,526 tracked vehicles**.

Overall System Methodology

six phases

1 Requirement analysis

Problem framed, smart-bump and camera-traffic literature reviewed, measurable objectives fixed.

2 System design

Architecture split into vision, relay and actuators; road and divider drawn in CAD; interfaces, pins and power allocated.

3 Build

Road fabricated, linkage 3D-printed, bump hinged to a servo, CV pipeline and three firmware images written.

4 Subsystem testing

Servo travel, the 20 cm threshold, the radio link, detection accuracy and signal timing each proved alone.

5 Integration

One measurement drives four actuators; priority ladder, common ground and separate actuator supply settled here.

6 Validation & documentation

833 automated checks, 69 recorded sessions, evaluation against O1-O7, then report, poster and demonstration.

Each phase closed only when its output could be **measured** — a servo angle against the physical plate, a speed against constructed ground truth, a command on the Serial Monitor.

Target Market & Marketing Strategy

who buys this

- Primary market — city councils and road authorities** retrofitting junctions where CCTV is already installed and a loop would mean digging.
- Secondary — emergency services, campuses and estates:** hospital and fire-station corridors, gated developments and universities that want calming without penalising ambulances.
- Route to market.** One instrumented junction as a paid pilot, a measured before-and-after report, then tender; kit sold near cost, analytics licensed per camera.
- Positioning and partners.** Cheaper than loops, no excavation, works on existing cameras, and the three subsystems sell separately — a council can start with the bump alone. CCTV integrators own the installed base; road-safety bodies and university pilots supply the independent evidence.

Environmental & Sustainability Considerations

retrofit, not rebuild

- No excavation.** The system retrofits onto cameras and poles that already exist, avoiding the road-cutting, lane closure and materials an inductive loop needs.
- Less idling.** Signals that answer the queue, and a bump that flattens for an emergency vehicle, both cut stop-start running — where urban fuel burn is worst.
- A small power budget.** Three microcontrollers and an LED strip on 5 V, 1 A; the GPU stays indoors and can serve several approaches.
- Materials chosen to be reworked.** Cardboard and PLA, printed rather than moulded parts, boards reused between builds — and adaptive calming wears road, brakes and tyres less than a permanent bump does.

Problems Encountered & Troubleshooting

— almost every rule in this system exists because something went wrong first, got measured, and got fixed

The LoRa receiver stayed silent

- A read of register 0x42 came back **0x00** — the module was not communicating at all. The SPI wiring, the NSS line, the supply and the radio settings were checked in turn and the firmware corrected. The receiver now detects and validates the E command.

The bump moved the wrong way

- Servo travel was opposite to the assumed 0°/90°. It was re-measured against the physical plate and recalibrated to **45° raised, 90° flat**.

Servos reset their own board

- Stall current was sagging the logic rail. Actuators moved to a separate 5 V 1 A supply with a common ground.

Empty road reported as HEAVY

- The index remembered the last speed it saw. Zero vehicles now means zero congestion, full stop.

Slow computer, slow cars

- A video was being timed by the wall clock. It now carries its own clock, one tick per frame.

29% of night traffic called "police"

- An unfair 14-against-15 vote. Made one-on-one, with the flashing beacon — not the livery — as the trigger.

The signal flickered between phases

- Yellow became committed to its full five seconds, plus a three-second minimum dwell.

Two programs, one COM port

- Windows refuses the second. The node owns the port and the dashboard asks it to act.

Sources of Error · Limitations · Future Work

— the honest edges of the prototype, and what each of them needs next

Sources of Error

- Calibration.** The homography tests on four clicked points and one typed dimension; a 5% error in that dimension scales every speed by 5% (±5% calibrated, ±25% automatic).
- Perspective and occlusion.** Distant vehicles occupy few pixels and their boxes jitter, so displacement is taken over 0.8 s and anything over 200 km/h is discarded as an identity swap.
- Ultrasonic physics.** A 15° beam, ~3 mm quantisation and a ~0.17%°C temperature dependence; soft or angled surfaces reflect poorly and a neighbouring object can trip the 20 cm gate.
- Open-loop actuators.** Gear backlash, horn stop and plate flex mean the commanded 45° is not exactly the physical angle — nothing measures the true position.
- Radio.** One character, no application-level acknowledgement; packet loss rises with distance and with interference on a shared 433 MHz band.
- Timing jitter.** USB serial and Windows scheduling add tens of milliseconds between a decision and a movement.
- Scale.** An 8 cm bump on a 360 mm road is geometrically but not dynamically similar to a real vehicle; timings here do not scale directly.
- Machine state.** The same benchmark re-run returned 16% slower with an identical vehicle yield — throughput describes a machine as much as a model.

Current System Limitations

- Congestion control from the ESP32 / dashboard over **A0 and A1 is not fully reliable**; the push-button path is.
- Vehicle-speed detection at the bump was not completed** — speed comes from the camera, not from the bump's own sensor.
- The LoRa link was formally demonstrated at a **bench distance of about 10 cm** only.
- The cardboard plate and micro servo suit a **scale prototype**; neither carries a vehicle load.
- No position feedback** on the bump or the divider — both are open-loop.
- No acknowledgement** is returned to the emergency vehicle, and the command carries no vehicle identity.
- No LCD** on the bump subsystem, so its state is visible only on the Serial Monitor.
- Untested under **real vehicle loads, rain or outdoor lighting**; a single approach, one camera, and no hand-labelled frames means no map is quoted.

Future Improvements

- Make the dashboard link deterministic — replace the A0 / A1 hint lines with a **framed, acknowledged UART or I²C message**.
- Add **speed measurement at the bump** with a two-gate ultrasonic pair, so calming can respond to speed as well as presence.
- Replace cardboard with **aluminium or steel plate** on a higher-torque actuator sized for a real wheel load.
- Fit **limit switches or an angle encoder** so the controller knows the plate angle instead of assuming it.
- Range-test the radio** properly — RSSI and packet loss logged from 10 m to the hundreds of metres LoRa is designed for.
- Add **authenticated, acknowledged emergency packages** with a rolling code and a vehicle identity, and echo confirmation to the driver.
- Build a **weatherproof IP65 enclosure** and repeat response-time and reliability runs (n ≥ 100) before any outdoor trial.
- Label a few hundred frames** for a real map, and port detection to an edge board so a laptop GPU is not required.

Conclusion

— what was aimed at, what was measured, and what is still open

Seven objectives were set and **seven were met**, with two stretch items met only in part and reported as such. A single camera and a laptop GPU measure real traffic at **27.1 fps** and return **±5% speed accuracy** once calibrated — 18, 36 and 72 km/h of constructed ground truth came back as 18.00, 36.00 and 72.00 (O1). Those speeds become one congestion index over **12,919 logged one-second samples**, across which mean speed falls monotonically from 42.2 to 10.0 km/h (O2), and that index drives a signal that cannot chatter: an exactly committed five-second yellow and a three-second dwell, proved against a synthetic clock (O3). Requiring a vehicle to be both liveried and flashing took night false alarms from **88 of 299 vehicles to zero** (O4).

On the physical prototype the bump raises to its calibrated **45°** for a vehicle detected within 20 cm and returns to **90° flat** on a validated LoRa E command, with emergency priority demonstrated over both congestion and automatic modes (O5); the movable divider answered all seven commands with LED and LCD state (O6), and the whole system is logged to CSV and XLSX and checked by **831 of 833 automated tests** (O7). What is not yet finished is stated just as plainly: the dashboard's analog hint path is unreliable, and the radio has only been demonstrated across a bench. The project therefore closes with a working, measured prototype — and a list of exactly what would have to be true before it belonged on a real street.

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Vision figures measured on an RTX 4050 laptop GPU, FP16 · methods and provenance in METRICS.md